

Sara Klaasen

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Education

PhD	ETH Zurich , Geophysics - Eidgenössische Technische Hochschule Zürich (ETH Zürich, Swiss Federal Institute of Technology Zurich) - ETH medal	Zurich, Switzerland 2020 – 2025
Msc	ETH Zurich , Geophysics	Zurich, Switzerland 2018 – 2020
Bsc	University College Roosevelt (UCR) , Liberal Arts and Sciences - Magna cum Laude - Major in Earth sciences, biology, mathematics and physics	Middelburg, Netherlands 2014 – 2017

Experience

Australian National University , Postdoctoral researcher Researching DAS projects in New Zealand. Imaging Taupo Volcano.	Canberra, Australia 2026 – present 1 year
ETH Zurich , Postdoctoral researcher	Zurich, Switzerland 2025 – 2026 1 year
Sport, Recreatie en Onderwijsvoorzieningen (SRO) , Admin Online administration	Haarlem, Netherlands 2017 – 2017 1 year
Provinciale Zeeuwse Courant (PZC) , Online editor - Online content management of the provincial newspaper - Writing and editing news articles	Vlissingen, Netherlands 2015 – 2017 2 years

Awards

ETH Medal The ETH Medal is given to outstanding doctoral theses (2k CHF). ETH Zurich	2026
Outstanding Student Presentation Award (OSPA) Seismological Society of America Photonics Meeting	2024

Grants

Swiss National Science Foundation (SNF) postdoc.mobility grant Grant to fund a 2-year postdoc position (136k CHF) Swiss National Science Foundation	2025
Women-in-STEM grant Grant for peer-mentoring group (2k CHF). Fix the Leaky Pipeline program, ETH Zürich	2021

VSbfonds grant

2018

Grant to support a Masters programme abroad (10k EUR).
VSbfonds

Research collaborations

Australian Research Council (ARC) Centre of Excellence in Optical Microcombs for Break-through Science (COMBS)

2026 – present

Associate investigator

www.combs.org.au

Seismicity on Venus: Prediction and Detection

2022 – 2024

International Space Science Institute (ISSI) international team.

ECR member

teams.issibern.ch/seismicityonvenus

Teaching experience

Data Analysis and Visualisation with Python in Earth Sciences

2022 – 2025

Bachelor course at ETH Zürich

Teaching assistant

Geophysical field lab

2021 – 2023

Bachelor fieldcourse at ETH Zürich

Teaching assistant

Dynamische Erde I

2020 – 2023

Annual ten minute lectures about current geophysical applications during the Bachelor course at ETH Zürich.

Guest lecturer

Research Output

Book Chapters

The leading edge of volcanology: Advances and innovations in monitoring and interpreting volcanic activity

2027

Book chapter submitted to "The encyclopedia of Volcanoes, Third Edition". Chapter has been accepted.

Michael P Poland, Elske de Zeeuw-van Dalfsen, Koki Aizawa,, Sara Klaasen, Fiona D'Arcy, Emma J Nicholson, Kye Anderson
(Elsevier)

Fiber optic sensing for volcano monitoring and imaging volcanic processes

2025

Book chapter in Modern Volcano Monitoring

Philippe Jousset, Gilda Currenti, Shane Murphy, Eva Eibl, Corentin Caudron, Lise Retailleau, Christopher Wollin, Sara Klaasen, Takeshi Nishimura, Andreas Fichtner, Zack Spica, Arnaud Lemarchand, Charlotte Krawczyk

doi.org/10.1007/978-3-031-86841-2_17 (Springer)

Journal Articles

Submarine seismicity monitoring with distributed acoustic sensing near Santorini and Kolumbo Volcano

2026

Sara Klaasen, Thomas Hudson, Sanchit Sachdeva, Paraskevi Nomikou, Andreas Fichtner

doi.org/10.26443/seismica.v5i1.1845 (Seismica)

- Seismic constraints on glacier density** 2025
Ariane Lanteri, Scott Keating, Lars Gebraad, Sara Klaasen, Marta Pienkowska-Cote, Olaf Eisen, Andrea Zunino, Kristín Jónsdóttir, Coen Hofstede, Dimitri Zigone, Andreas Fichtner
doi.org/10.1038/s41598-025-26440-z (Scientific Reports)
- The Seismogenic Thickness of Venus** 2025
Julia Maia, Ana-Catalina Plesa, Iris van Zelst, Richard Ghail, Anna Gülcher, Mark Panning, Sven Peter Näsholm, Barbara de Toffoli, Anna Horleston, Krystyna Smolinski, Sara Klaasen, Robert Herrick, Raphael Garcia
doi.org/10.1038/s41598-025-26440-z (Journal of Geophysical Research: Planets)
- Towards a widely applicable earthquake detection algorithm for fibreoptic and hybrid fibreoptic-seismometer networks** 2025
Thomas Hudson, Sara Klaasen, Olivier Fontaine, Conor Bacon, Kristin Jonsdottir, Andreas Fichtner
doi.org/10.1093/gji/ggae459 (Geophysical Journal International)
- An illustrated guide to: Distributed and integrated fibre-optic sensing in seismology** 2024
Andreas Fichtner, Fabian Walter, Patrick Paitz, Sara Klaasen, Daniel Bowden, Sebastian Noe, Nils Müller, Dominik Husmann, Jacques Morel
dx.doi.org/10.1016/j.eqsc.2024.09.006 (Earthquake Sciences)
- Seismic wave detectability on Venus using ground deformation sensors, infrasound sensors on balloons and airglow imagers** 2024
Raphael Garcia, Iris van Zelst, Taichi Kawamura, Sven Peter Näsholm, Anna Horleston, Sara Klaasen, Maxence Lefevre, Celine Marie Solberg, Krystyna Smolinski, Ana-Catalina Plesa, Quentin Brissaud, Julia Maia, Simon Stähler, Philippe Lognonné, Mark Panning, Anna Gülcher, Richard Ghail, Barbara de Toffoli
doi.org/10.1029/2024EA003670 (Earth and Space Science)
- Modelling uncertainty in P-wave arrival-times retrieved from DAS data: case-studies from 15 fibre optic cables** 2024
Emanuele Bozzi, Nicola Piana Agostinetti, Andreas Fichtner, Sara Klaasen, Arantza Ugalde, Biondo Biondi, S Yuan, Torsten Dahm, Marius Isken, Patrick Paitz, Fabian Walter, Alain Baird, Carlos Becerril, Takeshi Nishimura, Junzhu Shen, Tieyuan Zhu, Gilberto Saccorotti
doi.org/10.1093/gji/ggae364 (Geophysical Journal International)
- Complex spatial distribution of onset amplitude and waveform correlation: case studies from different DAS experiments** 2024
Emanuele Bozzi, Gilberto Saccorotti, Nicola Piana Agostinetti, Carlos Becerril, Andreas Fichtner, Sara Klaasen, Takeshi Nishimura, Junzhu Shen, Fabian Walter, Tieyuan Zhu
doi.org/10.4430/bgo00458 (Bulletin of geophysics and oceanography)
- Challenges in submarine fiber-optic earthquake monitoring** 2024
Jonas Igel, Sara Klaasen, Sebastian Noe, Paraskevi Nomikou, Konstantinos Korantzalos, Andreas Fichtner
doi.org/10.1029/2024JB029556 (Journal of Geophysical Research: Solid Earth)
- Subglacial volcano monitoring with fibre-optic sensing: Grímsvötn, Iceland** 2023
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristín Jónsdóttir, Lars Gebraad, Patrick Paitz, Andreas Fichtner
doi.org/10.30909/vol.06.02.301311 (Volcanica)

Fiber-optic observation of volcanic tremor through floating ice sheet resonance Andreas Fichtner, Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Patrick Paitz, Kristin Jonsdottir doi.org/10.1785/0320220010 (The Seismic Record)	2022
Distributed acoustic sensing in volcano-glacial environments — Mount Meager, British Columbia Sara Klaasen, Patrick Paitz, Nadja Lindner, Jan Dettmer, Andreas Fichtner doi.org/10.1029/2021JB022358 (Journal of Geophysical Research: Solid Earth)	2021
Detecting Seismic Events with Computer Vision: Applications for Fiber-Optic Sensing Solvi Thrastarson, Robert Torfason, Sara Klaasen, Patrick Paitz, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner doi.org/10.1002/essoar.10509693.1 (ESS Open Archive)	2021
First-Author Conference Proceedings	
Modelling months of deformation during the Sundhnúkur eruptions on the Reykjanes Peninsula, Iceland Sara Klaasen, Daniel Bowden, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Jack Maxwell, Meghan Miller, Andreas Fichtner (SSA optical seismology and the next era of seismic sensing)	2026
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (IAVCEI General Meeting)	2025
Full-Waveform Inversion of DAS data on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner (IAVCEI General Meeting)	2025
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (SSA General Meeting)	2025
Moment Tensor inversion with Full-Waveform Inversion and Distributed Acoustic Sensing on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (EGU Copernicus Meetings)	2024
Submarine volcano monitoring with fiber-optic cables at Santorini Sara Klaasen, Paraskevi Nomikou, Konstantinos Karantzas, Jonas Igel, Andreas Fichtner (AGU Annual Meeting)	2023
Going towards Full Waveform Inversion (FWI) with Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Jonas Igel, Andreas Fichtner (AGU Annual Meeting)	2023

Volcano monitoring and modelling using Distributed Acoustic Sensing (DAS) on a sub-glacial volcano: Grímsvötn, Iceland	2023
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Sebastian Noe, Andreas Fichtner (IUGG General Assembly)	
Monitoring glacially covered volcanoes with Fibre-Optic Sensing	2023
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (IAVCEI General Meeting)	
Volcanic Event Characterization with Fiber-Optic Sensing	2022
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (AGU Annual Meeting)	
Fibre-Optic Sensing for Volcano Monitoring on Grímsvötn, Iceland	2022
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (EGU General Assembly)	
Distributed Acoustic Sensing on Volcanoes: Grímsvötn and Fagradalsfjall, Iceland	2021
Sara Klaasen, Solvi Thrastarson, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Patrick Paitz, Andreas Fichtner (AGU Annual Meeting)	
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia	2021
Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner (SSA General Meeting)	
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia	2021
Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner (EGU General Assembly)	
Distributed Acoustic Sensing for the Exploration of the Mount Meager Volcanic Complex, British Columbia, Canada	2020
Sara Klaasen, Andreas Fichtner, Patrick Paitz, Jan Dettmer (AGU Annual Meeting)	

Invited Talks

Keck observatory workshop	2026
SSA optical seismology and the next era of seismic sensing, Kona (Pre-conference workshop)	
Frontiers in Fiber-Optics Sensing in Seismology	2025
IASPEI-IAGA, Lisbon (Conference session)	
ECR Keynote talk	2025
IAVCEI, Geneva (Conference keynote)	

Seminar Stanford University, San Francisco (Seminar)	2025
Seminar University of Washington, online (Seminar)	2024
Session: An innovative photonic vision of volcanoes and geothermal systems SSA Photonic Seismology, Vancouver (Session)	2024
Colloquium CSIRO, Perth (Seminar)	2023
Geophysics Weekly Meeting Australian National University, Canberra (Seminar)	2023
Seminar University of Canterbury, Christchurch (Seminar)	2023
Workshop on Distributed Acoustic Sensing AGU, online (Pre-conference workshop)	2021

Service

Convening conference sessions
IAVCEI, AGU

Peer-review
Seismica, SRL

Languages

Dutch
native

English
C2

German
B2

French
B1